Lecture 13

Special Distributions (Chapter 5)

Examples of discrete distributions and their properties

- **The Bernoulli and Binomial distributions**
  - Discussed in detail in previous lectures.

Summary ① Bernoulli: $X\begin{pmatrix} 1 & 0 \\ p & 1-p \end{pmatrix}$;

- $E(X) = p$; $\operatorname{var}(X) = p(1-p)$

② Binomial $X\begin{pmatrix} 0 & 1 & 2 & \dots & k & \dots & n \\ \cdot & \cdot & \cdot & \cdot & p_k & \cdot & \cdot \end{pmatrix}$

- $X \sim \text{Bin}(n,p)$ if $p_k = \mathbb{P}(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$.

- $X = X_1 + \dots + X_n$, where $X_1, \dots, X_n$ are independent, Bernoulli(p) random variables.
- $E(X) = np$; $\text{var}(X) = np(1-p)$.

VB ⊙ The binomial distribution is the distribution of the number of successes in n independent Bernoulli trials, which have (all of them) the same success probability p.

Reading Section 5.2

- Recall The Binomial distribution is associated with sampling schemes with replacement

A = red balls
B = blue balls
$A+B = n$ total balls

; $P = \frac{A}{A+B}$ = probability of selecting a red ball = probability of success.

If we draw balls from the box of n balls, with replacement (after a ball is drawn, we put it back), then X = the number of red balls drawn is a Binomial ($n = \frac{A}{A+B}$, n).

③ The Hypergeometric Distributions

- What happens if we want the distribution of successes out of n dependent Bernoulli trials?

Example (Sample without replacement)

- Box with: A red balls (success) and B blue balls.
- Select $n \leq A+B$ balls from the box, without replacement.
- $X_i = 1$ if the i-th ball drawn is red.
  $X_i = 0$ if the i-th ball drawn is blue.
- Are $X_1, \ldots, X_n$ independent? //

- $X_i = 1$ = "Success".

$$\mathbb{P}(X_2 = 1 \mid X_1 = 0) = \frac{A}{A+B-1}$$

$$\mathbb{P}(X_2 = 1 \mid X_1 = 1) = \frac{A-1}{A+B-1}$$

- Therefore, knowledge of $X_1 = 0$ or 1 changes the probability of what happens in the second draw. Moral: We cannot use the Binomial distribution for the distribution of the number of successes (red balls drawn).

Definition The random variable X that counts the number of successes in the scheme above is given by:

$$(10)\quad \mathbb{P}(X = k) = \frac{\binom{A}{k}\binom{B}{n-k}}{\binom{A+B}{n}},$$

for $\max(0, n-B) \le k \le \min(n, A)$, and zero otherwise.

• To see the restrictions on the range of k, Consider an example.

(i) $A = 5$, $B = 7$, $n = 4$

• In this case, the maximum possible number of red balls is $4 = \min(4, 5)$.

• The minimum number of red balls is

$$0 = \max(0, 4-7) = \max(0, -3)$$

(ii) $A = 5$, $B = 3$, $n = 4$.

• Let $X = \#$ of red balls. Then $n - X =$ the number of blue balls that are drawn. Of course:

$$n - X \leq B \iff n - B \leq X.$$ Therefore the minimum value of red balls that are drawn is always larger than $n - B = 4 - 3 = 1$ in our example. In general, since $X \geq n - B$ and $X \geq 0$, we require $X \geq \max\{0, n - B\}$.

• The distribution given by (10) is called the Hypergeometric distribution with parameters A, B and n.

$$E(X) = \frac{nA}{A+B}$$

$$\text{Var}(X) = \frac{nAB}{(A+B)^2} \cdot \frac{A+B-n}{A+B-1}.$$

Remark 1 If the sampling had been done WITH replacement, then $X \sim \text{Bin}(n, p = \frac{A}{A+B})$, so the mean would be the same, but the variances are very different, as

$$\text{Var}(X) = n \cdot \frac{AB}{(A+B)^2}.$$

Binomial

Remark 2 · Say we have an "infinite population", in which the number of successes and failures are extremely large, but $\frac{A}{A+B} \approx p$, $0 \leq p \leq 1$.

- Say we draw a fixed, finite, number n of elements from this population.
- Then: there is very little difference between drawing with or without replacement in terms of the distribution of successes. The Hypergeometric $(n, \frac{A}{A+B})$ and the Binomial $(n, p = \frac{A}{A+B})$ are very close. See Th. 5.3.4 for a perf.

However: This only happens in "almost infinite" populations!

The Poisson distribution

(1) Used to model the number of random arrivals in a given interval, assuming we know the rate of arrival in that interval, on average.

Example 1 At some store customers arrive at a rate of 4.5/hour, on average. If X = number of customers entering the store/hour, what is $\mathbb{P}(X \geq 5)$?

(2) Used to approximate the number of successes in n independent Bernoulli trials, when p is very small, and n is large, but $np \approx$ (some constant) λ. So, it approximates $Bin(n,p)$. for "rare events".

(1) # of misprints on a page.

(2) # of people living to 100 years in a community.

etc.

Definition (The Poisson(λ) distribution)

$$X \sim \begin{pmatrix} 0 & 1 & 2 & \dots & k & \dots \\ & & & & p_k & \dots \end{pmatrix},$$

where $p_k = \mathbb{P}(X=k) = e^{-\lambda} \cdot \frac{\lambda^k}{k!}$, and

λ is the parameter of the distribution.

Example (continued):

$\lambda = 4.5$ and

$$\mathbb{P}(X \geq 5) = 1 - \mathbb{P}(X \leq 4)$$

$$= 1 - \sum_{k=0}^{k=4} e^{-4.5} \cdot \frac{(4.5)^k}{k!}.$$

Properties: (a) $\boxed{E(X) = \mathrm{Var}(X) = \lambda}$

(b) (The Poisson approximation to the Binomial)

If $X \sim Bin(n,p)$ and $n \cdot p = \lambda$, when $n \to \infty$

then, if $Y \sim Poisson(\lambda)$,

$$\mathbb{P}(X=k) \underset{\text{As } n\to\infty}{\approx} \mathbb{P}(Y=k).$$

(See th. 5.4.5 for a rigorous statement).

Example 2 An item produced by a machine is defective with prob. $p = 0.1$. Compute the probability that a sample of 10 items will contain at most 1 defective item.

Solution ① If the probability model is $Bin(10, 0.1)$,

then $\mathbb{P}(X \le 1) = \binom{10}{0}(0.1)^0 \cdot (0.9)^{10} + \binom{10}{1}(0.1)^1 (0.9)^9$

$\approx \boxed{0.7361}$

② If the probability model is $Poisson(\lambda = 10 \cdot 0.1 = 1)$,

then $\mathbb{P}(X \le 1) = e^{-1} \cdot \frac{1^0}{0!} + e^{-1} \cdot \frac{1}{1!} = 2 \cdot e^{-1} \approx \boxed{0.7349}$,

very close to ①!

Example 3 The expected number of typos on a page of a certain magazine is 0.2. What is the probability that the page you read will have no typos?

Solution : $X \sim \text{Poisson}(\lambda = 0.2)$,

$$\mathbb{P}(X=0) = e^{-0.2} \cdot \frac{(0.2)^0}{0!} = e^{-0.2}$$

Theorem If $X_1, \ldots, X_n$ are independent Poisson random variables with respective parameters $\lambda_1, \ldots, \lambda_n$, then

$X_1 + \ldots + X_n$ is $\text{Poisson}(\lambda)$,

where $\lambda = \lambda_1 + \ldots + \lambda_n$.

Example 1 continued

Assume: 1) X = # of customers in a first hour $\sim$ Poisson(4.5).

2) Y = # of customers in the second hour $\sim$ Poisson(4.5).

3) Y independ. of X.

Then: $X+Y \sim$ Poisson ($4.5+4.5=9$) can model the # of customers coming to the store within the first 2 hours.

$\mathbb{P}(X+Y \geq 12) = 0.1970$ (see Example 5.4.3 on how to use the Table in the book).

- Poisson process → NOT required.

Not required for exam, but good to know!

The Negative Binomial Distribution

- Count the number of failures until a fixed number of successes occur. What is the distribution of the corresponding random variable?

Example: A machine produces good and bad parts.

$X_i = 1$, if i is defective ("A success")
$X_i = 0$, if i is good.

The X_i's are independent of each other and, for all i, $\mathbb{P}(X_i = 1) = p$.

- Observe the production process until you see 4 defectives.
- X = the number of good parts observed before you see 4 defectives.
- Find p.f. of X.

Definition (Negative Binomial with parameters r and p)

- Infinite sequence of independent Bernoulli trials, each with probability of success p.
- X = the number of failures that occur before r successes are observed has p.f:

$$\mathbb{P}(X=k) = \binom{r+k-1}{k} p^r \cdot (1-p)^k,$$

for $k = 0, 1, 2, \ldots$

- Therefore, in our example, X is a Negative Binomial with p and $r=4$.
- In general:

$$E(X) = \frac{r(1-p)}{p},$$

$$\text{Var}(X) = \frac{r(1-p)}{p^2}.$$

Definition (The Geometric distribution with parameter p = The Negative Binomial distribution with parameters p and $r=1$).

$$\mathbb{P}(X=k) = \begin{cases} p(1-p)^k, & k=0,1,2,\cdots \\ 0, & \text{otherwise} \end{cases}$$

Counts the number of failures until the first success.

$$E(X) = \frac{1-p}{p}, \quad \text{var}(X) = \frac{1-p}{p^2}.$$